

EOB Extractor — Dependency Manifest

Date: 2026-06-08
Runtime: Node.js 20.x
Language: TypeScript ~5.9.3

1. Production Dependencies

Package	Version	Purpose	Notes
@aws-sdk/client-bedrock-runtime	^3.1014.0	Bedrock InvokeModel API	InvokeModelCommand, BedrockRuntimeClient
@aws-sdk/client-dynamodb	^3.1014.0	DynamoDB low-level client	Used in DynamoDBDocumentClient.from()
@aws-sdk/client-s3	^3.1014.0	S3 GetObject / PutObject	PDF read; quarantine move
@aws-sdk/client-sfn	^3.1014.0	Step Functions StartExecution	eob-trigger
@aws-sdk/client-sns	^3.1014.0	SNS Publish	eob-lookup-insurance notifications
@aws-sdk/client-sqs	^3.1014.0	SQS SendMessage	eob-store-result review queue
@aws-sdk/lib-dynamodb	^3.1014.0	DynamoDB Document client	PutCommand, QueryCommand
aws-cdk-lib	^2.243.0	CDK constructs	Infrastructure only
constructs	^10.5.0	CDK constructs base	Infrastructure only
ulid	^3.0.2	ULID generation	Correlation IDs, extraction IDs
zod	^4.3.6	Runtime schema validation	eob-extraction.schema.ts — uses zod/v4 import path

2. Development Dependencies

Package	Version	Purpose
@types/aws-lambda	^8.10.161	TypeScript types for Lambda events (SQSEvent)
@types/jest	^30	Jest type definitions
@types/node	^24.10.1	Node.js type definitions
aws-cdk	2.1112.0	CDK CLI
esbuild	^0.27.4	Lambda bundler (via CDK NodejsFunction)

Package	Version	Purpose
jest	^30	Test framework
ts-jest	^29	Jest TypeScript transformer
ts-node	^10.9.2	TypeScript execution (CDK bin)
typescript	~5.9.3	TypeScript compiler
vitest	^4.1.1	Unit test runner (primary)

3. AWS Service Dependencies

Service	Version / Config	Dependency Type
Lambda runtime	NODEJS_20_X	Runtime
Lambda architecture	ARM_64	Infrastructure
Step Functions	STANDARD type	Orchestration
DynamoDB	Standard (not global)	Storage
Bedrock	bedrock-2023-05-31 Messages API	AI/ML
KMS	Customer-managed CMKs	Security
SQS	Standard queues (not FIFO)	Messaging

4. Bedrock Model Dependencies

Model ID	Chain	Tier	Notes
us.anthropic.claude-haiku-4-5-20251001-v1:0	CLASSIFY_CHAIN[0], EXTRACT_CHAIN[2]	On-demand	Cross-region inference profile
us.anthropic.claude-sonnet-4-20250514-v1:0	CLASSIFY_CHAIN[1], EXTRACT_CHAIN[1]	On-demand	Cross-region inference profile
us.anthropic.claude-sonnet-4-6	EXTRACT_CHAIN[0]	On-demand	Cross-region inference profile

All models accessed via cross-region inference profiles (`us.anthropic.*`). IAM policies grant `bedrock:InvokeModel` for foundation model ARNs in `us-east-1` , `us-east-2` , `us-west-2` and inference profile ARNs in the deployment region.

5. External Data Dependencies

Resource	Owner	Access	Criticality
Insurance-Arbitration-contacts DynamoDB table	Platform team	Read + conditional write	HIGH — pipeline lookup step fails without it
bucket-specialops / bucket-specialops-sandbox S3 bucket	Platform team	Read (clickup/* prefix)	HIGH — source of all PDFs

Neither resource is managed by the EOB Extractor CDK stack. Changes (deletion, rename, IAM policy changes) require coordination with the platform team.

6. CDK Context Dependencies

The stack requires the following CDK context values at deploy time:

Context Key	Default	Required
environment	sandbox	No
bucketName	bucket-specialops-sandbox	No
contactsTable	Insurance-Arbitration-contacts	No
costCenter	engineering	No

7. Vulnerability Scanning Notes

- All AWS SDK packages are locked at `^3.1014.0` — pin and update together to avoid SDK version skew
- `aws-cdk-lib` and `constructs` are infrastructure-only and not bundled into Lambda artifacts
- `zod` v4 uses the `zod/v4` import path — verify compatibility if upgrading from v3
- `ulid` v3 API may differ from v2 — `ulid()` function import unchanged